# Predicting Wikipedia Pageviews of UNESCO World Heritage Sites

## 1 Research Question and Objectives

To what extent can real-world characteristics of UNESCO World Heritage properties, particularly their geographical accessibility, historical background, and classification, predict their Wikipedia pageviews?

I initially expected Wikipedia-specific characteristics, such as article length, the number of linked articles, and language availability, to be strong predictors of pageviews. However, I was particularly interested in investigating whether real-world characteristics could also help explain differences in online attention.

Therefore, this project compares machine learning models to explore whether factors such as heritage category, geographical location, inscription history, and nearby urban population can help predict differences in Wikipedia pageviews.

## 2 Data Collection and Integration

This project combines data from several sources to investigate whether real-world characteristics of UNESCO World Heritage properties can help predict their Wikipedia pageviews. The datasets were integrated using UNESCO property identifiers and geographical coordinates.

Table 1 provides an overview of the datasets used in the project.

Table 1: Overview of the datasets used in the project.

| Dataset | Source | Purpose |
|---|---|---|
| UNESCO World Heritage List | UNESCO DataHub | Property characteristics |
| World Cities Database | Simplemaps | Urban population and city distances |
| Airport Database | OurAirports | Airport accessibility |
| Wikipedia article mappings | Wikidata SPARQL | Linking properties to Wikipedia articles |
| Wikipedia pageviews | Wikimedia API | Prediction target |

### 2.1 UNESCO World Heritage Data

The main dataset, `whc001`, was obtained from the UNESCO World Heritage List on the UNESCO DataHub. It contains 1,273 properties included on the World Heritage List.

The dataset provides information about each property, including its inscription date, geographical location, classification, endangered status, and whether it extends across national borders. The following variables were used or investigated for the initial predictive model:

- **ID:** The UNESCO identifier used to link properties to their corresponding Wikipedia article mappings.
- **Date inscribed:** The date the property was added to the World Heritage List, used to calculate the number of years since inscription.

- **Category:** Classifies properties as cultural, natural, or mixed.
- **ISO Codes:** The country codes associated with the property.
- **Coordinates:** The geographical coordinates used to calculate nearby city populations and other accessibility measures.
- **Danger:** Indicates whether the property is included on UNESCO's List of World Heritage in Danger.

To calculate geographical accessibility measures, the coordinates of each World Heritage property were compared with city locations from the Simplemaps World Cities Database and airport locations from OurAirports.

### 2.2 Wikipedia Data

Wikipedia article mappings were collected using the Wikidata SPARQL endpoint. These mappings identify Wikipedia articles associated with UNESCO World Heritage properties across six language editions: English, Russian, Chinese, Arabic, French, and Spanish. Monthly pageview data was subsequently retrieved using the Wikimedia Pageviews API.

Wikipedia articles were identified for 1,008 of the 1,273 World Heritage properties, representing approximately 79.2% coverage. A distinction was made between UNESCO World Heritage properties and Wikipedia articles. While each UNESCO property represents one observation in the predictive dataset, a property may be associated with multiple Wikipedia articles, including articles about individual components of a property.

Pageview data was collected for January 2026. This period was selected to provide a consistent observation window and to allow pageviews to be compared with historical characteristics such as the date of inscription.

The API retrieval process takes several hours and includes request delays, retry handling, and periodic progress saving to support reliable data collection.

## 3 Data Exploration and Preprocessing

Before preprocessing, I examined missing values, variable types, and relationships between the available variables in the UNESCO dataset. The original dataset contained 1,273 properties and 54 fields, including geographical information that was partly redundant. To avoid unnecessary complexity and overlapping information, I selected the variables most relevant to the research question.

I excluded image and video fields because the project focuses on predicting Wikipedia pageviews using property characteristics rather than analysing multimedia content. I also retained only the English names and excluded descriptions. For the Wikipedia data, however, I collected pageviews across six languages to capture a broader range of online attention.

The dataset contained missing values. One notable finding was that the *Danger list* field was missing for approximately 95% of properties. Further investigation revealed that its missingness was related to the *Danger* variable: the field was populated for properties on the Danger List but missing for properties not on it. Since these variables conveyed overlapping information, I retained only *Danger* for the analysis.

The category variable distinguishes cultural, natural, and mixed properties. More detailed categorisation was considered, but the final analysis contained only 989 observations because some properties did not have identified Wikipedia articles. With this relatively limited sample size, a more detailed categorisation could increase the risk of overfitting.

## 3.1 Feature Engineering

### 3.1.1 Continent

Continents were derived from the ISO country codes provided by UNESCO using the Python package `country_converter`. After inspecting the results, I identified properties with missing or multiple continent classifications. Three properties had missing continent values, while four were initially classified as spanning multiple continents. These cases were investigated individually, and continent assignments were manually corrected where appropriate. For transboundary properties spanning multiple continents, both relevant continents were retained in the final feature representation.

### 3.1.2 Geographical Accessibility and Remoteness

To investigate whether the geographical remoteness of a World Heritage property is related to its Wikipedia pageviews, I used two additional datasets: the Simplemaps World Cities Database, containing city locations and population estimates, and OurAirports, containing airport locations, types, and scheduled-service information.

Using these datasets and the geographical coordinates of the World Heritage properties, I calculated several measures of geographical accessibility. Before performing the calculations, records with missing or invalid geographical coordinates were excluded. City records with missing population estimates or negative population values were also removed.

Distances between World Heritage properties and nearby cities or airports were calculated using the Haversine formula. This formula estimates the great-circle distance between two geographical coordinates on the Earth's surface, providing a straight-line distance in kilometres. It is defined as:

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1)\cos(\phi_2)\sin^2\left(\frac{\Delta\lambda}{2}\right) \tag{1}$$

$$d = 2R\arcsin(\sqrt{a}) \tag{2}$$

where $\phi_1$ and $\phi_2$ represent the latitudes, $\lambda_1$ and $\lambda_2$ represent the longitudes, and $R$ is the Earth's radius, approximated at 6,371 km. The resulting distances were used to calculate the following accessibility measures:

- **Nearby urban population:** The total population of all cities located within 50 km and 100 km of each property, calculated by summing the population estimates of cities within the respective radius.
- **Distance to the nearest city:** The straight-line distance to the closest city, regardless of its population.
- **Distance to larger cities:** The distance to the nearest city with a population of at least 100,000 and 500,000 inhabitants, respectively.
- **Distance to the nearest large airport:** The distance to the closest airport classified as a large airport with scheduled commercial service.
- **Nearby large airports:** The number of large airports with scheduled service located within 100 km of each property.

These measures were designed to capture different aspects of geographical accessibility, ranging from proximity to urban populations to access to larger cities and international air transport.

Since several geographical accessibility measures capture related aspects of remoteness, each candidate feature was evaluated individually rather than including all geographical measures simultaneously. Each feature was added to the baseline feature set, and its predictive performance was compared using five-fold cross-validation. MAE was used as the primary selection criterion. This approach provided a straightforward way to identify which geographical measure offered the greatest additional predictive value without unnecessarily increasing model complexity. Mean absolute error (MAE) was used as the primary selection metric. Since several measures captured similar aspects of geographical accessibility, I aimed to avoid including multiple overlapping features in the final model. Each candidate feature was therefore evaluated separately to identify the measure that provided the greatest improvement in predictive performance.

#### 3.1.3 Years Since Inscription

Years since inscription was calculated using the year 2026 minus the year in which a property was inscribed. Properties inscribed in 2026 were excluded to avoid using information that would not have been available at the time of prediction.

### 3.2 Merging the UNESCO Data with Wikipedia Pageviews

After feature engineering, the original geographical columns and the inscription date were dropped. The UNESCO data was then merged with the January 2026 Wikipedia pageview data using the UNESCO property ID.

### 3.3 Target Variable

The target variable, total Wikipedia pageviews in January 2026, was constructed by aggregating the pageviews of all identified Wikipedia articles associated with each UNESCO World Heritage property. Pageviews were summed across the six selected language editions: Arabic, Chinese, English, French, Russian, and Spanish. This resulted in a single pageview count for each property. The resulting target therefore represents the pageviews of identified and mapped articles rather than necessarily all Wikipedia attention related to a property.

The original distribution of pageviews was strongly right-skewed, with most World Heritage properties receiving relatively few views and a small number receiving substantially more. To reduce this skewness, a logarithmic transformation was applied using $\log(1 + \text{pageviews})$. The addition of 1 ensures that observations with zero pageviews can also be included. This transformation reduces the influence of properties receiving exceptionally high numbers of views and makes the target variable more suitable for regression modeling. Figure 1 illustrates the distribution of Wikipedia pageviews before and after the logarithmic transformation.

January 2026 was selected as the observation period to provide a consistent measure of Wikipedia attention across all properties. This also allows years since inscription to be calculated relative to 1 January 2026, allowing properties inscribed in 2025 to be included while ensuring that the feature is based on information available before the observation period.

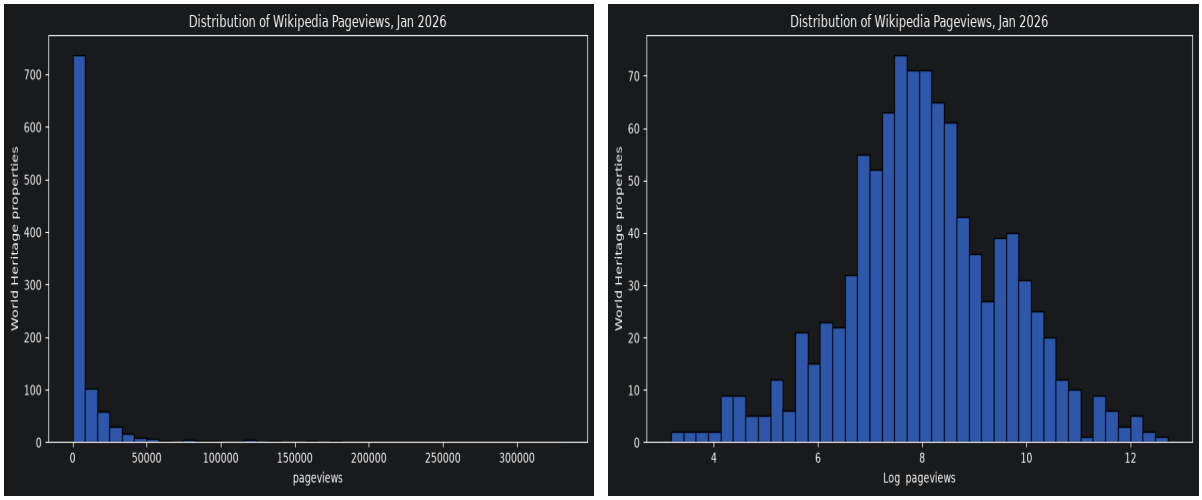



Figure 1: Distribution of Wikipedia pageviews before and after log transformation.

## 4 Model Development and Validation

To investigate whether geographical accessibility helps predict Wikipedia attention, several geographical features were evaluated. These included nearby city population, distance to cities of different population sizes, and proximity to large scheduled airports. Each candidate feature was added individually to the baseline feature set and evaluated using five-fold cross-validation. Mean absolute error (MAE) was used as the primary selection metric, while root mean squared error (RMSE), $R^2$, median absolute error, and explained variance provided additional measures of predictive performance.

Given the relatively limited number of observations, five-fold cross-validation was used instead of a separate train-test split. This approach makes more efficient use of the available data, as each observation is used for both training and validation across the five folds. In each iteration, four folds are used for training and the remaining fold for validation.

The results for Linear Regression are presented in Table 2. City population within 50 km achieved the lowest MAE (1.1528), representing a small improvement over the baseline model (1.1545). This feature was therefore selected for the Linear Regression model.

The Random Forest feature comparison is shown in Table 3. Distance to the nearest city with at least 100,000 inhabitants performed best, achieving an MAE of 1.1668 compared with 1.1807 for the baseline. This feature was retained for the Random Forest model.

Table 2: Geographical feature comparison for Linear Regression.

| Feature | MAE | RMSE | $R^2$ | Median AE | Explained variance |
|---|---|---|---|---|---|
| Baseline | 1.1545 | 1.5005 | 0.0671 | 0.9192 | 0.0671 |
| City population within 50 km | 1.1528 | 1.4969 | 0.0717 | 0.9084 | 0.0717 |

Table 3: Geographical feature comparison for Random Forest.

| Feature | MAE | RMSE | $R^2$ | Median AE | Explained variance |
|---|---|---|---|---|---|
| Baseline | 1.1807 | 1.5158 | 0.0480 | 0.9433 | 0.0480 |
| Nearest city (100,000+ inhabitants) | 1.1668 | 1.5065 | 0.0597 | 0.9217 | 0.0597 |

Two regression models were subsequently evaluated: Linear Regression and Random Forest. Linear Regression was selected as an interpretable model, while Random Forest was included to

investigate whether a model capable of capturing nonlinear relationships and feature interactions could improve predictions. A Dummy Regressor served as a reference point by predicting the mean target value. All models were evaluated using the same cross-validation folds and performance metrics.

The Random Forest model was further tuned by varying `min_samples_leaf`, which determines the minimum number of training observations required in each leaf. The number of trees and other hyperparameters were kept fixed to isolate the effect of leaf size. As shown in Fig. **??**, model performance generally improved as the minimum leaf size increased from smaller values, reaching its best results at 30. Beyond this point, performance deteriorated slightly. Therefore, a leaf size of 30 was retained for the final comparison, as it achieved the lowest MAE and the highest $R^2$ among the tested configurations.

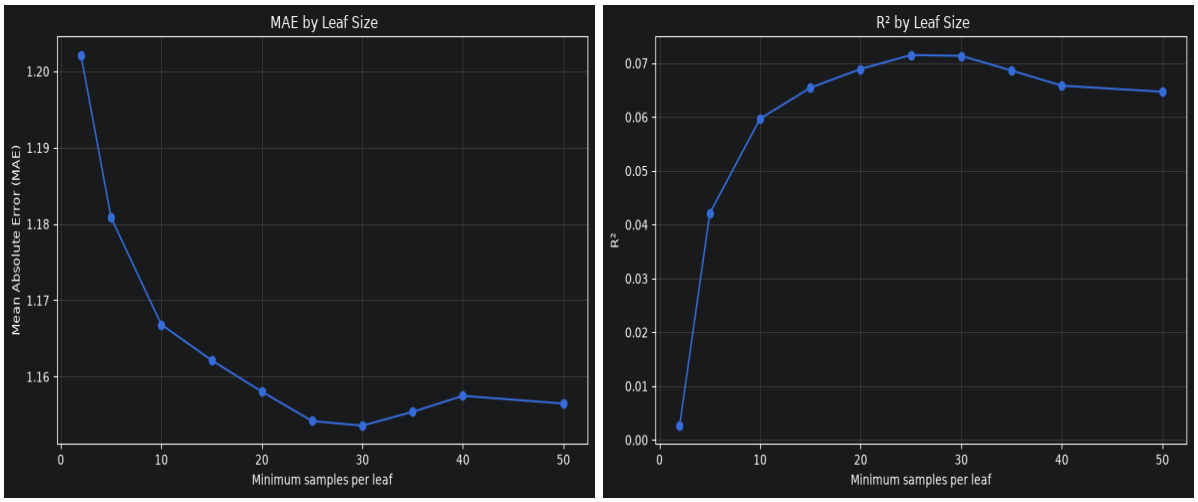



Figure 2: Random Forest MAE and $R^2$ by Leaf Size, Feature: nearest city 100K distance km.

## 4.1 Model Interpretation

## 4.2 Model Interpretation

The fitted models provide insight into which characteristics contain the most predictive information about Wikipedia attention. The Linear Regression coefficients and Random Forest feature importances are presented in Table 4.

Years since inscription was the most prominent feature in both models. It had the largest standardised coefficient in Linear Regression (0.4157) and the highest feature importance in Random Forest (0.4862). The positive coefficient indicates that older inscriptions are associated with higher predicted Wikipedia pageviews. This may reflect the time these properties have had to become established in public awareness and accumulate historical recognition.

Geographical accessibility also contributed to the predictions. City population within 50 km had a positive standardised coefficient in Linear Regression (0.1252), while distance to the nearest city with at least 100,000 inhabitants was the second-most important feature in Random Forest (0.2816). These results highlight the relevance of the surrounding urban environment and proximity to population centres when predicting Wikipedia attention.

Asia also showed a positive coefficient in Linear Regression (0.2692) and relatively high feature importance in Random Forest (0.1703). In comparison, the remaining continent indicators and the Cultural and Natural category variables had smaller contributions. Danger had a small negative coefficient in Linear Regression (-0.0357) and zero feature importance in Random Forest, indicating a limited contribution to the predictions.

Overall, inscription history and geographical context were the most informative characteristics among those considered. However, these results represent model-specific associations rather than causal relationships, and the importance measures of the two models are not directly comparable. Furthermore, both models explained only around 7% of the variation in log-transformed Wikipedia pageviews. This suggests that, although the selected features capture some patterns in Wikipedia attention, much of its variation remains unexplained.

Table 4: Linear Regression coefficients and Random Forest feature importances.

| Feature | Linear Regression coefficient | Random Forest importance |
|---|---|---|
| Years since inscription | 0.4157 | 0.4862 |
| Continent: Asia | 0.2692 | 0.1703 |
| Geographical feature | 0.1252 | 0.2816 |
| Category: Cultural | 0.0856 | 0.0045 |
| Category: Natural | 0.0430 | 0.0088 |
| Danger | -0.0357 | 0.0000 |
| Continent: America | -0.0355 | 0.0184 |
| Continent: Oceania | -0.0237 | 0.0000 |
| Continent: Europe | -0.0164 | 0.0243 |
| Continent: Africa | -0.0058 | 0.0059 |

Note: The geographical feature is city population within 50 km for Linear Regression and distance to the nearest city with at least 100,000 inhabitants for Random Forest. Linear Regression coefficients are standardised.

## 5 Results and Discussion

The final model comparison is presented in Table 5. Linear Regression achieved an MAE of 1.1528 and an $R^2$ of 0.0717. The tuned Random Forest, using a minimum leaf size of 30, achieved an MAE of 1.1536 and an $R^2$ of 0.0714. Both models outperformed the Dummy Regressor, which achieved an MAE of 1.2031 and an $R^2$ of -0.0020.

The two regression models performed almost identically, with Linear Regression achieving marginally better MAE and $R^2$. Despite its ability to capture nonlinear relationships and feature interactions, the tuned Random Forest offered no meaningful predictive improvement. This may indicate that the available features contain limited nonlinear patterns or that important determinants of Wikipedia attention are missing from the dataset. Given their similar performance, Linear Regression is preferable due to its greater interpretability.

Overall, the results indicate that the available heritage and geographical characteristics provide some predictive signal, but explain only a small proportion of the variation in Wikipedia attention. The limited difference between Linear Regression and Random Forest suggests that the simpler linear model is sufficient for the predictive performance achieved in this project.

Table 5: Final model performance comparison using five-fold cross-validation.

| Model | MAE | RMSE | $R^2$ | Median AE | Explained variance |
|---|---|---|---|---|---|
| Linear Regression | 1.1528 | 1.4969 | 0.0717 | 0.9084 | 0.0717 |
| Random Forest (`min_samples_leaf` = 30) | 1.1536 | 1.4971 | 0.0714 | 0.8912 | 0.0714 |
| Dummy Regressor | 1.2031 | 1.5551 | -0.0020 | 0.9515 | -0.0020 |

## 6 Limitations

This project has four main limitations. First, pageviews were aggregated across six Wikipedia language editions, which do not cover all languages and may introduce geographical biases in the representation of certain properties. Second, the article-matching process may miss articles related to a UNESCO property, such as those about individual components or related features. Their pageviews are therefore excluded from the target, potentially underestimating the property's total Wikipedia attention. Third, the target covers only January 2026, so the results may reflect seasonal patterns or temporary events rather than typical levels of attention.

Finally, selecting features and tuning hyperparameters using the same cross-validation results may lead to optimistic performance estimates. A separate held-out test set could provide a more independent evaluation, but I chose to retain as many observations as possible given the limited dataset size. Future work could address these limitations by including more language editions, improving article matching, analysing multiple months, and evaluating the model on a held-out test set when more data are available.

## 7 Conclusion

This project investigated whether real-world characteristics of UNESCO World Heritage properties can help predict their Wikipedia pageviews. In particular, it examined the predictive value of historical characteristics, heritage classifications, geographical location, and accessibility.

The results indicate that these characteristics contain some predictive information about Wikipedia attention. Years since inscription emerged as the most important feature in both regression models, while geographical accessibility also contributed to predictive performance. However, the relatively low $R^2$ values indicate that most of the variation in Wikipedia pageviews remains unexplained.

Linear Regression and Random Forest achieved almost identical performance, suggesting that the more complex model did not provide a meaningful predictive advantage over the simpler linear approach. Both models outperformed the Dummy Regressor, although the overall improvement remained modest.

Ultimately, the findings suggest that real-world characteristics alone are insufficient to accurately predict Wikipedia attention. Future research could investigate whether combining these characteristics with Wikipedia-specific features and pageview observations from multiple months would improve predictive performance.